

Matteo Pettenò



PROFILE

AI Researcher and Software Engineer specializing in deep learning, audio processing and controllable generative models. Formerly a Marie Skłodowska-Curie Doctoral Fellow within the Privacy for Smart Speech Technology network (jointly hosted by EURECOM and Ruhr-Universität Bochum), I published research on voice anonymization alongside my prior academic work in controllable music generation. I hold an M.Sc. in Music & Acoustic Engineering (Politecnico di Milano), a B.Sc. in Information Engineering (University of Padova), and multi-year industry experience as a Full-Stack Developer, DevOps Engineer, and Software Architect. **I am fully available for full-time roles in AI/ML and software engineering.**

SKILLS

- Python, C++, MATLAB, GLSL, JAVA, SQL, JS, CSS
- Keras, Tensorflow, PyTorch
- NumPy, SciPy, scikit-learn
- Apache Beam, Apache Airflow, Apache Spark
- GCP, AWS, CI/CD, Docker, LXC, Proxmox
- Tone.js, Three.js, Svelte, Vue.js, Firebase, Flask, Spring, PWA, Workbox, Hugo
- JUCE, Supercollider
- librosa, FMP Notebooks
- Logic Pro, Ableton Live, Reaper, Ardour
- COMSOL, REW
- gdb, OWASP ZAP, ghidra
- Figma

EDUCATION

Ph.D. Candidate in PSST (Marie Curie Fellow)

Digital Security Department, EURECOM
IKA, Ruhr-Universität Bochum (RUB)

09/2025 - 02/2026

Sophia Antipolis, France
Bochum, Germany

- *Privacy for Smart Speech Technology* (PSST) joint doctoral training programme
- **Research focus:** Disentangled representations for selective attribute suppression

Master's Degree in Music and Acoustic Engineering

DEIB, Politecnico di Milano (PoliMi) - *Graduated Cum Laude*

09/2021 - 10/2024

Milan, Italy

- Relevant Courses: Machine Learning, Computer Music, Sound Analysis Synthesis and Processing, Creative Programming and Computing, Musical Acoustics, Electronics and Electroacoustics, Computer Security
- Thesis: *Latent Space Regularization via Normalizing Attribute Transformations for Symbolic Music Generation*

Bachelor's Degree in Information Engineering

DEI, University of Padua (UNIPD)

09/2013 - 07/2021

Padua, Italy

- Relevant Courses: Algorithms for Engineering, Systems and Models, Control systems, Electronics, Telecommunications
- Thesis: *Evaluation of the performance of commercial STT and NER services applied to digitized oral sources*

PUBLICATIONS

S. Chandra, **M. Pettenò**, N. Evans et al. "Evaluating voice anonymisation using similarity rank disclosure", in Proc. of the *Odyssey - The Speaker and Language Recognition Workshop*, Lisbon, Portugal, June, 23-26, 2026

M. Pettenò, A. I. Mezza and A. Bernardini, "Conditional Diffusion As Latent Constraints for Controllable Symbolic Music Generation", in Proc. of the *26th Conference of the International Society for Music Information Retrieval (ISMIR)*, Daejeon, Korea, Sept. 21-25, 2025

M. Pettenò, A. I. Mezza and A. Bernardini, "On the Joint Minimization of Regularization Loss Functions in Deep Variational Bayesian Methods for Attribute-Controlled Symbolic Music Generation", in Proc. of the *33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, Sept. 8-12, 2025.

R. B. Luzietti et. al, "FONTI 4.0: Evaluating Speech-to-Text Automatic Transcription of Digitized Historical Oral Sources". *Proceedings of the Eighth Italian Conference on Computational Linguistics CliC-it*, 2021

PROFESSIONAL EXPERIENCE

Full Stack Developer

ccelera s.r.l (Arsenalia Group) - Via Lepetit, 8, 20124

11/2021 - 08/2023

Milan, Italy

- Platforms: SAP Hybris Commerce
- Customers: Bonfiglioli, Cellularline, PegPerego, Metal Work

DevOps Engineer

Walit s.r.l - Via Dandolo, 25/B, 31100

09/2019 - 02/2021

Treviso, Italy

- Platforms: Google Cloud Platform (GCP), Gitlab, Flask, OWASP ZAP

System Integration Engineer

Alpenite Ltd - 38 Craven Street, WC2N 5NG

01/2019 - 07/2019

London, UK

- Platforms: Mulesoft, RabbitMQ, FTP
- Customers: Stella McCartney

Full Stack Developer

Alpenite s.r.l (Arsenalia Group) - Via delle Industrie, 27/7, 30175

01/2017 - 01/2019

Venice, Italy

- Platforms: SAP Hybris Commerce
- Customers: Kering Eyewear

RESEARCH INTERESTS

- Deep Learning
- Representation Learning
- Speech Processing
- Privacy Preservation
- Speaker Anonymization
- Deepfake Detection
- Music Information Retrieval
- Audio Generation
- AI-Assisted Music Composition
- Music Understanding
- Hearing Aids

LANGUAGES

Italian: Mother tongue

English: Fluent (C1)

French: Base (A1)

HOBBIES

- **Music:** Guitarist and mixing/recording engineer for my band @venus_in.cancer.
- **Outdoors:** Nature enthusiast. Experienced in solo and group wild bikepacking journeys and mountain trekking.
- **DIY Projects:** Creative builder with a resource-driven mindset. Built a custom wood shed, handcrafted a balance board, upcycled an antique family bicycle into a coaster-brake bike, experimented with homebrewing and many more.
- **Sports:** Lifelong basketball player.
- **Gardening:** Dedicated gardener maintaining my home garden.

PERSONAL PROJECTS

Unbaised

03/2025 - 11/2025

Co-Creator/Lead Developer

Remote, Venice (Italy) / San Francisco (USA)

Keywords: python, docker, gcp, expo, tavily, eleven labs.

Engineered an AI platform dynamically generating news and podcasts tailored to user-defined bias metrics. Automated daily multi-source news ingestion, parameter-conditioned article synthesis and audio podcast generation.

Homelab

2022 - Present

Self-Hosted Homelab & Smart Home

Keywords: proxmox, *arr, jellyfin, nextcloud, caddy, pi-hole, wg, gitea, HA, zigbee

Architected and maintaining a Proxmox VE hypervisor running self-hosted media, storage, networking and home automation services.

Around the Job

2023 - Present

Full Stack Developer & Creator

[github](#)

Keywords: vue.js, firebase, firestore, leaflet

Advent of Code

2022 - Present

Challenge Solutions & API

[link](#)

Keywords: python, flask, render, coding-challenge

RESEARCH PROJECTS

Evaluating voice anonymisation using similarity rank disclosure

2026

In Proc. of the Odyssey 2026 - The Speaker and Language Recognition Workshop

Keywords: voice anonymisation, privacy, evaluation

Conditional Diffusion as Latent Constraints for Unconditional Symbolic Music Generation Models

2025

In Proc. of the 26th Conference of the ISMIR

[github](#)

Keywords: symbolic music, attribute-controlled generation, diffusion models, latent constraints

On the Joint Minimization of Regularization Loss Functions in Deep Variational Bayesian Methods for Attribute-Controlled Symbolic Music Generation

2025

In proc. of the 33rd EUSIPCO

[github](#)

Keywords: symbolic music, attribute-controlled generation, power transforms

Do Unconditional Deep Generative Models Spontaneously Learn How to Encode Human-Interpretable Musical Attributes?

2023

Music and Acoustics Engineering Capstone course in MS.

[github](#)

Keywords: variational autoencoders, latent space topological structure

Evaluation of the performance of commercial STT and NER services applied to digitized oral sources

2021

Thesis in Information Engineering BS

[github](#)

Keywords: speech-to-text, named-entity-recognition, gcp, aws

CREATIVE PROJECTS

Ego

2023

Creative Programming & Computing course in MS

[github](#)

Keywords: three.js, glsl, svelte, mediapipe, max4live, tone.js

Pulseq - Fractal Sequencer

2022

Advanced Coding Tools and Methodologies course in MS

[github](#)

Keywords: fractal sequencer, web app, svelte, tone.js, glsl